PHY 614 (2011)

Problem Set # 4

Problem 1 (1.10.1)

$$\int dx\, \delta(ax) f(x) = \frac{1}{a}\int dy\, \delta(y) f(y/a) = \frac{1}{a} f(0)$$

$$\Rightarrow \qquad \delta(ax) = \frac{1}{a}\delta(x)$$

Problem 2 (1.10.2)

Consider a zero of $f(x)$ at $x = x_i$
Then assuming that $f(x)$ has a Taylor series expansion around this point

$$f(x) = (x - x_i) f'(x_i) + \frac{1}{2}(x - x_i)^2 f''(x_i) + \cdots$$

$\delta(f(x))$ is nonzero only near $x = x_i$. Thus for each such zero we may write

$$\delta(f(x)) = \delta\big((x - x_i) f'(x_i)\big)$$

$$= \frac{1}{|f'(x_i)|}\delta(x - x_i)$$

The full result is then obtained by summing over all zeros

Problem 3

Consider, for any function $f(x)$

$$\int_{-L}^{L} dx \, \frac{d\Theta(x)}{dx} f(x)$$

$$= \left[\Theta(x) f(x)\right]_{-L}^{L} - \int_{-L}^{L} dx \, \Theta(x) f'(x)$$

$$= f(L) - \int_{0}^{L} dx \, f'(x)$$

$$= f(L) - \left[f(x)\right]_0^L = f(0)$$

Furthermore

$$\frac{d\Theta}{dx} = 0 \quad \text{everywhere except at } x = 0$$

and at $x = 0$ $d\Theta/dx = \infty$.

$$\Rightarrow \quad \delta(x) = \frac{d\Theta}{dx}$$

Problem 4

(a) $\langle \phi | P \psi \rangle = \int dx\, \phi^*(x) P \psi(x)$

$= \int_{-\infty}^{\infty} dx\, \phi^*(x) \psi(-x)$

$= \int_{-\infty}^{\infty} dx\, \phi^*(-x) \psi(x).$

$= \int_{-\infty}^{\infty} dx\, (P\phi(x))^* \psi(x)$

$= \langle P\phi | \psi \rangle = \langle \phi | P^\dagger | \psi \rangle.$

Thus $P = P^\dagger$

(b) $P^2 \psi(x) = P\psi(-x) = \psi(x)$

$\Rightarrow P^2 = \mathbb{I}$

Since $P = P^\dagger$ $\quad P^2 = PP^\dagger = \mathbb{I}$

$\Rightarrow P$ is unitary

(c) Since $P^2 = 1$ the eigenvalues must be ± 1

(d) For some $\psi(x)$

$\psi_\pm = \frac{1}{2}(\psi(x) \pm \psi(-x))$

Satisfy

$P\psi_\pm(x) = \pm \psi_\pm(x)$

Problem 5:

(a) $$\langle \Phi | L\psi \rangle = \int_0^{2\pi} d\varphi \, \Phi^*(\varphi) \, -i\frac{d\psi}{d\varphi}$$

$$= \left[-i\,\Phi^*(\varphi)\psi(\varphi) \right]_0^{2\pi} + i\int_0^{2\pi} d\varphi \, \frac{d\Phi^*}{d\varphi}\psi$$

Since $\Phi(\varphi) = \Phi(\varphi+2\pi)$
$\psi(\varphi) = \psi(\varphi+2\pi)$ the first term vanishes

$\Rightarrow$

$$\langle \Phi | L\psi \rangle = \int d\varphi \left(-i\frac{d\Phi}{d\varphi}\right)^* \psi = \langle L\Phi | \psi \rangle$$

$$= \langle \Phi | L^\dagger | \psi \rangle \quad \Rightarrow \quad L = L^\dagger$$

(b) $$e^L = \sum_{n=0}^{\infty} \frac{1}{n!} L^n$$

$$(e^L)^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!} (L^n)^\dagger = \sum \frac{1}{n!} (L^\dagger)^n$$

$$= e^{L^\dagger} = e^L$$

(c) Eigenvalue eqn is

$$-i\frac{d\psi}{d\varphi} = \alpha\psi$$

$$\Rightarrow \quad \psi = A e^{i\alpha\varphi}$$

Since $\psi(\varphi+2\pi) = \psi(\varphi)$

$$\alpha = n \qquad n = 0, \pm 1, \pm 2 \dots$$

(d) Normalize $\quad A = \frac{1}{\sqrt{2\pi}}$

Problem 6

(a) $a^\dagger = \frac{1}{\sqrt{2}}(X^\dagger - iK^\dagger) = \frac{1}{\sqrt{2}}(X - iK)$

(b) $[a, a^\dagger] = \frac{1}{2}[X + iK, X - iK]$

$= \frac{i}{2}[K, X] - \frac{i}{2}[X, K]$

$= \frac{i}{2}(-i) - \frac{i}{2}(i) = 1$

(c) $\langle\lambda| X |\lambda\rangle = \frac{1}{\sqrt{2}}\langle\lambda| a + a^\dagger |\lambda\rangle$

$= \frac{1}{\sqrt{2}}\langle\lambda| \lambda^* + \lambda |\lambda\rangle = \sqrt{2}\,\mathrm{Re}\,\lambda$

where we used

$\langle\lambda| a^\dagger = \lambda^* \langle\lambda|$

$\langle \cancel{x} | K |\lambda\rangle = \frac{1}{\sqrt{2}i}\langle\lambda| a - a^\dagger |\lambda\rangle$

$= \frac{1}{\sqrt{2}i}(\lambda - \lambda^*) = \sqrt{2}\,\mathrm{Im}\,\lambda$

(d) In this representation

$a = \frac{1}{\sqrt{2}}\left(x + \frac{d}{dx}\right)$

Thus

$\frac{1}{\sqrt{2}}\left(x + \frac{d}{dx}\right)\psi_\lambda(x) = \lambda\,\psi_\lambda(x)$

~~To~~ solve this

$\frac{1}{\psi_\lambda}\frac{d\psi_\lambda}{dx} = (\sqrt{2}\lambda - x)$

$\ln\psi_\lambda = (\sqrt{2}\lambda x - \frac{1}{2}x^2) + \text{const}$

$\Rightarrow$

$\psi_\lambda(x) = A e^{-\frac{1}{2}x^2 + \sqrt{2}\lambda x}$

This is a gaussian packet.
Normalization

$$|A|^2 \int dx\, \psi_\lambda^*(x)\psi_\lambda(x) = |A|^2 \int dx\, e^{-x^2 + \sqrt{2}(\lambda+\lambda^*)x}$$

$$= |A|^2 \int dx\, e^{-(x^2 - 2\sqrt{2}\,\mathrm{Re}\lambda\, x + 2(\mathrm{Re}\lambda)^2)}\, e^{2(\mathrm{Re}\lambda)^2}$$

$$= |A|^2 \int dx\, e^{-(x-\sqrt{2}\,\mathrm{Re}\lambda)^2}\, e^{2(\mathrm{Re}\lambda)^2}$$

$$= |A|^2 \sqrt{\pi}\, e^{2(\mathrm{Re}\lambda)^2}$$

Thus may choose

$$A = \frac{1}{\pi^{1/4}} e^{-(\mathrm{Re}\lambda)^2}$$

(e) $$\langle\lambda|x^2|\lambda\rangle = \langle\lambda|\tfrac{1}{2}(a+a^\dagger)^2|\lambda\rangle$$

$$= \tfrac{1}{2}\langle\lambda|a^2|\lambda\rangle + \tfrac{1}{2}\langle\lambda|(a^\dagger)^2|\lambda\rangle + \tfrac{1}{2}\langle\lambda|aa^\dagger|\lambda\rangle + \tfrac{1}{2}\langle\lambda|a^\dagger a|\lambda\rangle$$

$$= \tfrac{1}{2}\langle\lambda|a^2|\lambda\rangle + \tfrac{1}{2}\langle\lambda|a^{\dagger 2}|\lambda\rangle + \tfrac{1}{2}\langle\lambda|\lambda\rangle + \langle\lambda|a^\dagger a|\lambda\rangle$$

$$= \tfrac{1}{2}(\lambda+\lambda^*)^2 + \tfrac{1}{2}$$

for normalized $|\lambda\rangle$

Thus

$$\langle\lambda|x^2|\lambda\rangle - (\langle\lambda|x|\lambda\rangle)^2$$

$$= \frac{1}{2}(\lambda+\lambda^*)^2 + \frac{1}{2} - \frac{1}{2}(\lambda+\lambda^*)^2 = \frac{1}{2}$$

Similarly

$$\langle\lambda|K^2|\lambda\rangle = -\frac{1}{2}\langle\lambda|(a-a^\dagger)^2|\lambda\rangle$$

$$= -\frac{1}{2}\left[\langle\lambda|a^2|\lambda\rangle + \langle\lambda|a^{\dagger 2}|\lambda\rangle - \langle\lambda|aa^\dagger + a^\dagger a|\lambda\rangle\right]$$

$$= -\frac{1}{2}(\lambda^2 + \lambda^{*2} - 2\lambda\lambda^*) + \frac{1}{2}$$

$$= -\frac{1}{2}(\lambda-\lambda^*)^2 + \frac{1}{2}$$

$\Rightarrow$

$$\langle\lambda|K^2|\lambda\rangle - \langle\lambda|K|\lambda\rangle^2 = \frac{1}{2}$$